

AN2DL - Second Challenge Report

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1 Introduction

Regarding this four-class image classification problem, we employ a patch-based approach where images are subdivided into smaller regions. These patches are processed using **Multiple Instance Learning (MIL)** aggregation techniques to produce a slide-level prediction.

2 Problem Analysis

The dataset consists of pathology images categorized into four subtypes: **Luminal A**, **Luminal B**, **HER2(+)**, and **Triple negative**, includes 691 training images and their mask files.



Figure 1: The distribution of classes in the dataset

The dataset exhibits class imbalance, with the **Luminal B** subtype being the most prevalent, **Triple negative** subtypes are underrepresented.

We are trying to extract multiple patches from a pathology image.

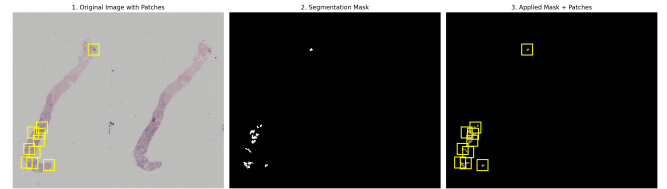


Figure 2: Example of extracted patches from a pathology image (0000)

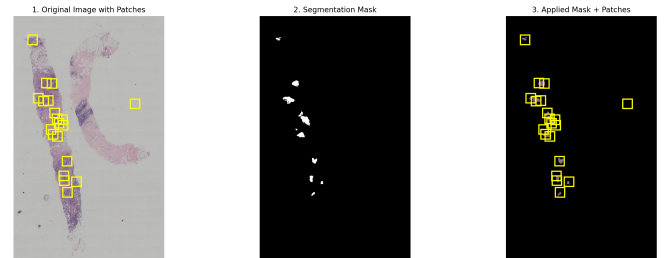


Figure 3: Example of extracted patches from a pathology image (0002)

3 Method

3.1 Noisy image recognition

We observed that some images in the dataset are noisy or corrupted (Orcs and Slime), which can negatively impact model performance. To address this, we used two layers of defense:

- **Color Check:** Histology images are never green. If an image has significant green pixels (common for orcs/slimes), it is immediately trashed.
- **CLIP Classification:** Use `openai/clip-vit-large-patch14` to identify and filter out these noisy images from the training set. We create text prompts like "a histology image" and "a green slime histology image" to classify images. If an image is classified as "green slime histology image" with high confidence, it is marked as noisy.

Finally, we removed **101** noisy images from the training set, get **590** clean images for training.

3.2 Data Augmentation

To enhance model robustness and generalization, we applied various data augmentation techniques during training, including

- Random rotations and flips
- Color jittering (brightness, contrast, saturation adjustments)
- Mixup augmentation

3.3 Dual-stream MIL Network

We use a Dual-Stream Multiple Instance Learning (MIL) architecture designed to integrate local morphological details with global spatial context. As illustrated in 4, the network processes a image through two parallel streams: a *Local Stream* (Bag of Instances) and a *Global Stream* (Contextual View), fusing their representations for final classification.

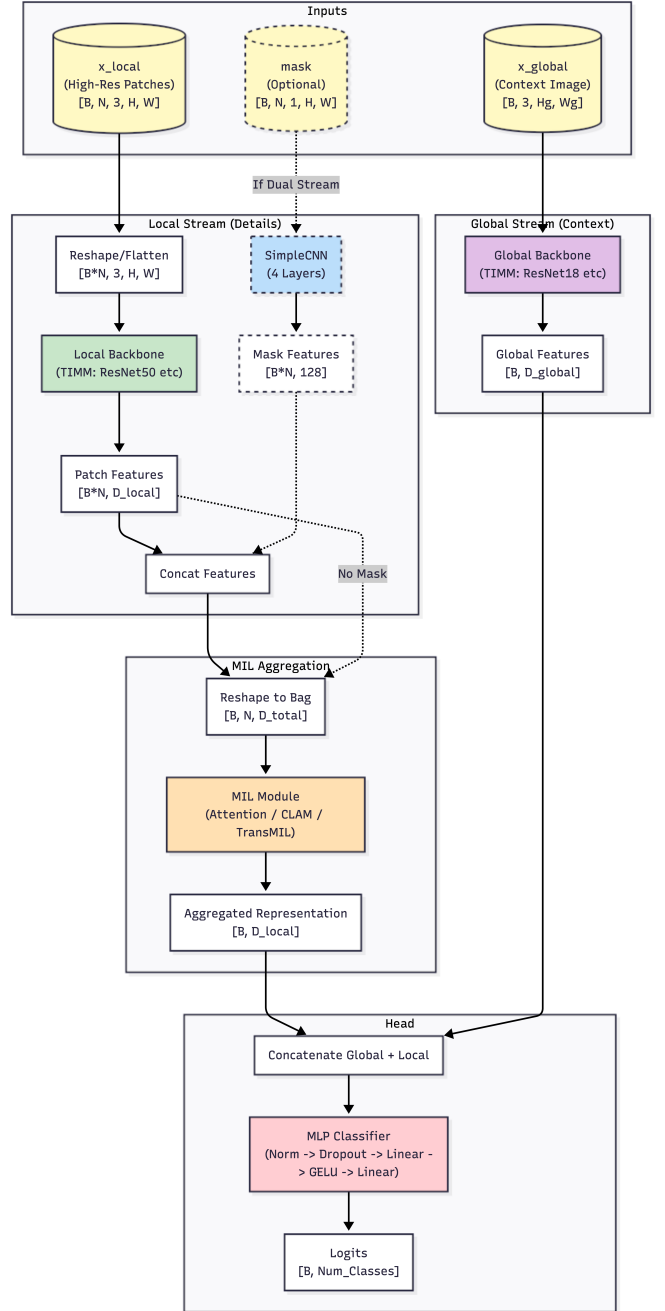


Figure 4: Dual-Stream MIL Architecture

Feature Extraction Each patch x_i is processed independently by a shared backbone encoder f_θ (e.g., ResNet50), yielding a raw feature vector. To reduce dimensionality and enforce feature selection, we apply a bottleneck B_{local} consisting of a linear projection, LayerNorm, GELU activation, and Dropout:

Optional Mask Fusion If binary segmentation masks m_i are provided, they are processed by a

lightweight 4-layer CNN, g_ϕ . The resulting mask embeddings are concatenated with the patch features:

MIL Aggregation To obtain a permutation-invariant representation of the entire bag, we employ an aggregation function $\mathcal{A}(\cdot)$. While the architecture supports CLAM and TransMIL, the general attention-based aggregation is defined as: where a_i represent learnable attention scores computed via a Gated Attention mechanism (Ilse et al., 2018) or a Transformer encoder, ensuring the model focuses on diagnostically relevant patches.

To capture macro-architectural patterns often lost in patch-based tiling, we introduce a global stream. A downsampled view of the ROI or WSI, X_{global} , is processed by a secondary backbone q_ψ (e.g., ResNet18). Similar to the local stream, this is projected via a bottleneck B_{global} :

The final slide-level representation $\mathbf{Z}$ is obtained by concatenating the aggregated local features and the global context vector: This fused vector is passed to a multi-layer perceptron (MLP) classifier with aggressive dropout and GELU activation to predict the class probability distribution $\hat{Y}$:

The model is trained using Cross-Entropy loss with support for Mixup augmentation and label smoothing to improve generalization.

4 Experiments

For your experiments, you might want to present your results in tables. Here’s an example of a wide table comparing different models:
For more specific measurements, you might use a narrower table:

Table 2: An example of table. Best results may be highlighted in **bold**.

Time [μ s]	Distance [mm]
22 $\pm$ 4	8 $\pm$ 1
17 $\pm$ 3	7 $\pm$ 1
15 $\pm$ 3	6 $\pm$ 1
13 $\pm$ 2	5 $\pm$ 1
10 $\pm$ 2	4 $\pm$ 1
8 $\pm$ 2	3 $\pm$ 1
5 $\pm$ 1	2 $\pm$ 1
37 $\pm$ 1	1 $\pm$ 1

You can also include figures to visualise your results:



Figure 5: Example figure showing [describe what the figure shows]

Reference figures using like:“As shown in Figure 5...”

5 Results

Present your main findings here. You might want to:

- Compare your results with baselines
- Highlight key achievements using **bold text**
- Explain any unexpected outcomes

6 Discussion

In this section, analyse your results critically. Consider:

- Strengths and weaknesses
- Limitations and assumptions

7 Conclusions

Summarise your work and discuss potential future directions. This is where you can:

- Restate main contributions
- Suggest improvements
- Propose future work

Table 1: An example of wide table. Best results are highlighted in **bold**.

Model	Accuracy	Precision	Recall	ROC AUC
VGG18	72.20 ± 3.06	94.95 ± 0.52	86.95 ± 0.55	80.16 ± 0.81
Custom Model	27.71 ± 3.19	75.70 ± 1.07	55.75 ± 2.16	36.60 ± 1.26
ResNet18	89.24 ± 2.38	95.54 ± 0.49	93.43 ± 1.30	91.68 ± 0.71